

Walchand College of Engineering, Sangli

Computer Science & Engineering

Third Year

Course: Computer Algorithm Lab

Assignment No. 4

Greedy Method

1. An airline wants to load the maximum possible number of checked-in bags into a cargo hold that has a strict weight capacity, before the flight departs.

Input sample:

Cargo Hold Capacity = 3000 kg

Bag weights (kg) = [22, 18, 40, 15, 60, 33, 27, 50, 12, 45, 38, 29, 55, 19, 42, 31, 24, 48, 36, 20, 58, 16, 47, 25, 34, 41, 13, 52, 28, 37, ...] (150 bags total)

Implement the Container Loading problem to maximize the number of bags loaded without exceeding the cargo hold capacity.

2. A freelance marketplace has 50 open tasks posted by clients, each with a deadline (expressed as a slot number, 1–10 days) and a payout. A freelancer can complete at most one task per day and, once a task's deadline passes, it can no longer be taken up. Select and schedule a subset of tasks that maximizes the freelancer's total payout.

Input sample:

Tasks = T1(deadline=2, payout=450), T2(1,300), T3(3,220), T4(2,600), T5(1,150), T6(4,700), T7(3,90), T8(2,500), T9(4,260), T10(1,800), ... (50 tasks total)

Slots Available = 10

Implement the Job Sequencing with Deadlines problem to maximize total payout.

3. A cargo ship with a fixed hold capacity needs to be loaded with different commodities, each with a total available quantity and a value. Bulk commodities such as crude oil or grain can be loaded in any fraction, while manufactured goods such as machinery or vehicles must be loaded as a whole or not at all. Maximize the total value of cargo loaded.

Input sample:

Capacity = 1000 tons

Commodities = CrudeOil(qty=300, value=900), Coal(qty=250, value=700), Machinery(qty=150, value=650, indivisible), Grain(qty=200, value=500), Steel(qty=180, value=620, indivisible), Chemicals(qty=100, value=450), Vehicles(qty=120, value=800, indivisible), Textiles(qty=90, value=300), Electronics(qty=60, value=700, indivisible), Fertilizer(qty=140, value=400), ... (60 commodities total)

Implement the Fractional Knapsack problem to maximize the total value loaded within the ship's capacity.

4. A hospital network needs to merge patient record files arriving from four branch clinics into one master file. Files can only be merged two at a time, and the cost of merging two files equals the sum of their sizes. Determine the merge order that minimizes the total merge cost.

Input sample:

Round 1 (Branch A): [35, 110, 180, 8]

Round 2 (Branch B): [25, 12, 240, 85, 55]

Round 3 (Branch C): [95, 65, 30, 50, 80, 40]

Round 4 (Branch D): [480, 290, 140, 20, 65]

(File sizes in MB, 20 files total)

Implement the Optimal Merge Pattern problem to minimize the total cost of merging all files into one.

5. A satellite transmits telemetry data from 20 onboard instrument channels back to ground control. Each channel has a transmission frequency (occurrences per orbit) and a priority factor (1–5); channels that are both frequent and high-priority should receive the shortest possible codes to minimize transmission time and power usage.

Input sample:

Channels with (Frequency x Priority): Temperature(380), Pressure(560), Voltage(300), Gyro(150), Thruster(110), AttitudeControl(190), SolarPanel(240), BatteryLevel(420), Communication(500), Radiation(70), Magnetometer(60), StarTracker(200), ReactionWheel(170), FuelLevel(130), Antenna(90), Camera(210), GPS(85), Accelerometer(65), DataBus(150), Timing(100)

Expected Output:

Shortest Codes -> Pressure='0', Communication='10', Temperature='110', ...

Longest Codes -> Magnetometer='1111110', Radiation='1111111'

Average Code Length ~ 5.2 bits

Implement Huffman Coding to generate optimal prefix codes for the telemetry channels.